A clear, specific title stating what you investigated

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2026-11-09

Executive summary. 4–6 sentences, written for a non-specialist. State the research question, the data, the method, and the key finding, translating any statistics into plain language. Write this last, once the report is complete.

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Code and data: all analysis for this report is reproducible from the repository at https://github.com/USER/REPO.

1 Introduction

Election polls have a vital role in examining voter preferences and influencing expectations about electoral results. However, their stated precision does not automatically guarantee

accuracy. In the Journal of the American Statistical Association, Shirani-Mehr et al. (2018) discovered that observed polling errors were around double the implied conventional margins of error, emphasising that polling uncertainty extends beyond random sampling variability. This prompts a question about whether polling characteristics can explain differences in the accuracy of polls.

The 2012 and 2016 US presidential elections offer a valuable basis for exploring this question. Particularly, state-level polling in the election year 2016 demonstrated substantial errors, with Kennedy et al. (2018) noting a mean absolute error of 5.1 percentage points. The journal article states that late swings in voter preferences and insufficient adjustments for educational attainment were key factors that led to such errors.

These findings further motivate an examination into whether polling timing, sample size, and survey methodology are associated with the accuracy of polling results. Hence, this report investigates the research question: **To what degree are polling timing, sample size and methodology associated with the accuracy of state-level presidential polls for the 2012 and 2016 elections.** More specifically, it explores:

1. Whether polling accuracy is associated with days prior to Election Day

2. Whether election year, sample size and methodology explain variation in polling errors once timing is accounted for

Polling accuracy is measured as the absolute error between polling and actual Democratic-Republican voting margins. Multiple linear regression models are utilised as a means to explore these relationships.

2 Method

2.1 Data

This report utilises state-level presidential polling data from the US election years 2012 and 2016. This data was retrieved from HuffPost Pollster via the pollstR R package on 10 January 2017 as part of the rOpenGov project. Official election results were obtained from the MIT Election Data and Science Lab's US President 1976-2024 dataset, accessed via the Harvard Dataverse. Following the data preparation, the unit of observation in the final dataset is an individual state-level poll, which can be identified using election year, state and poll ID.

Initial data cleaning and standardisation was completed previously in the Exploratory Data Analysis Report. These datasets can be found in the data folder within the GitHub Repository (add in link). For the following analysis, party candidate support percentages for Democratic and Republican were combined into a single observation per poll. Polling margins were measured as the difference between Republican and Democratic support (Democratic – Republican). A similar calculation was used for actual election outcomes, with electoral margins

being measured using each party's share of total state votes. Polls were matched to election observations based on state and year.

The response variable, absolute polling error, calculates the absolute difference between polling and election margins in terms of percentage points. Explanatory variables to be used include:

- days_prior: the number of days between poll's end date and the relevant Election Day
- sample_size_log: the logarithm of the sample size for the poll
- mode: polling methodology (Phone, Internet and Mixed/Other)
- year: election year – either 2012 or 2016

As noted, sample size was logarithmically transformed given there may be a potentially non-linear relationship with polling error. Additionally, the mode variable was consolidated from 8 categories into 3 groups: Phone, Internet and Mixed/Other. This consolidated reduced the number of different categories with minimal observations, enabling a more effective comparison between polling modes.

Any observations that were incomplete or had invalid values for the variables required for regression modelling were excluded, enabling consistent samples across all models. The final dataset consists of 2436 observations. The descriptive statistics for this dataset is displayed in Table 2. The additional data cleaning and preparation is fully documented in Section 4.

Table 1: Descriptive statistics of numerical variables

Variable	Mean	SD	Median
abs_error	5.16	4.20	4.13
days_prior	77.12	95.51	38.00
sample_size	837.50	464.25	716.00

Table 2: Descriptive statistics of numerical variables

Election year	Mode	No. of polls	Percent (%)
2012	Phone	939	82.8
2012	Internet	119	10.5
2012	Mixed / Other	76	6.7
2016	Phone	531	40.8
2016	Internet	578	44.4
2016	Mixed / Other	193	14.8

2.2 Statistical methods

Multiple linear regressions using ordinary least squares was utilised to explore associations between polling characteristics and absolute polling error. The use of this method allowed for the relationship between polling timing and error to be examined, while accounting for other explanatory variables.

A total of four nested regression models were constructed to address the central research question. Model 1 investigates the association between polling time and absolute error. Model 2 includes an additional explanatory variable for year to account for differences between 2012 and 2016 elections. Model 3 adds log-transformed sample size, and Model 4 further includes polling mode to investigate methodological differences once adjusting for other variables. This final model considers absolute polling error as a function of timing, election year, sample size and mode.

In these models, election year and polling mode were treated as categorical variables, with 2012 and Phone as the respective reference categories. The coefficients of the models were interpreted as changes in expected absolute polling error (in percentage points), accounting for other predictors included in the given model.

Model assumptions including linearity, constant variance, and residual normality were assessed through residual versus fitted and normal Q-Q plots. Additionally, influential observations were also investigated using standard regression diagnostics. Potential dependence between polls from the same state or pollster was considered a possible limitation of the assumption of independence. Given that absolute errors are non-negative and possibly right-skewed, assumption checking is especially relevant. The analysis remains observational, meaning any relationships should not be considered as causal. Detailed model specifications, diagnostics, and R code is fully documented and can be found in Section 4.

3 Results

As displayed in Figure 1, absolute polling error tends to increase with greater number of days remaining before Election Day. This suggests that polls carried out earlier appear to deviate more from the actual election results. The 2016 polls also indicate greater errors compared to 2012, but there is moderate variation for both election years. These preliminary relationships are further investigated in a formal manner using multiple linear regression, with estimated models showcased in Table 3.

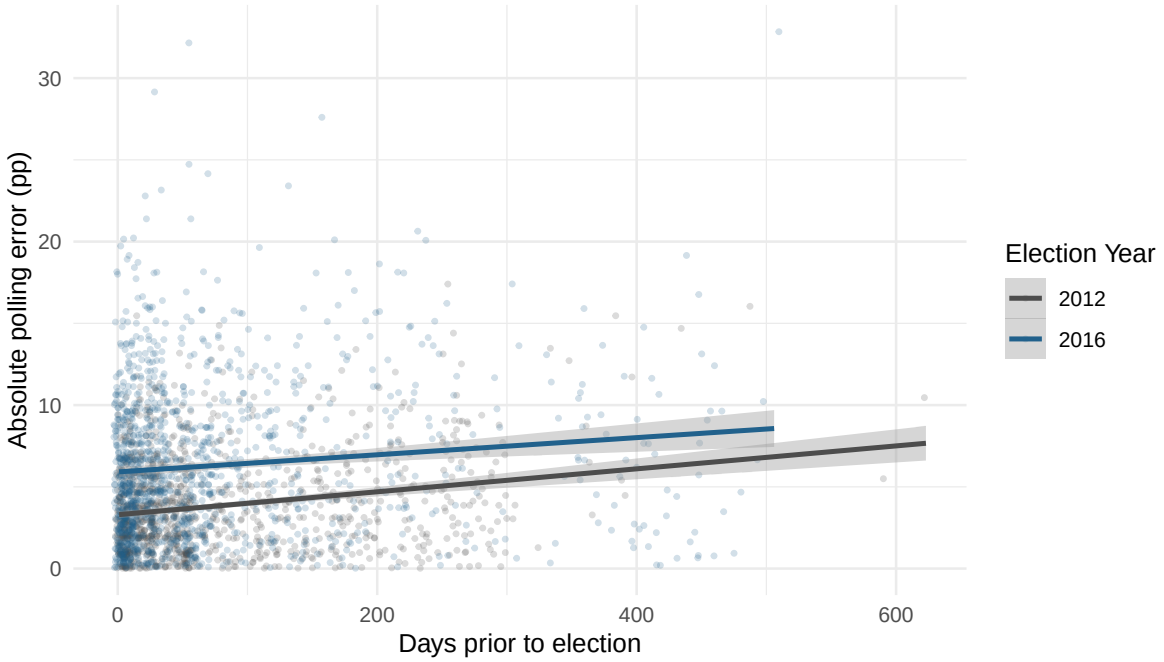


Figure 1: Absolute polling error in Democratic-Republican vote margins (pp) against number of days the poll was conducted prior to the 2012 and 2016 elections. Each point indicates an individual poll, with grey representing election year 2012, and navy indicating 2016. Solid lines are fitted linear trends using OLS for each election year, and shaded bands represent 95% confidence interval. Figure suggests that polls conducted further from the election date tend to have larger errors, while errors are mostly greater in 2016. The model doesn't account for sample size and polling mode, and larger errors could possibly indicate changes in voter preferences nearing election day.

The regression estimates displayed in Table 3 offer further evidence of an association between polling timing and absolute polling error. In Model 1, each additional day before the elections is associated with a 0.006 percentage point (0.006%) increase in absolute error. Over 100 days, this corresponds to an increase of around 0.556 percentage points. Across all four models, the relationship between poll timing and error remains positive and statistically significant, after accounting for sample size, mode and election year. This pattern indicates that polling timing is consistently associated with accuracy. In saying this, the magnitude of the relationship is modest.

Model 2 introduces the election year, and substantially improves explanatory power. This can be seen from the R squared value increasing from 0.016 to 0.103. This means that including election year increases the amount of variation that can be explained by the model by approximately 8.662 percentage points. The estimated coefficient for 2016 is positive and

Table 3: Multiple linear regression exploring association between polling characteristics and absolute polling error for 2012 and 2016 US presidential elections.

	Model 1	Model 2	Model 3	Model 4
Intercept	4.735***	3.378***	7.712***	7.597***
	(0.109)	(0.136)	(1.148)	(1.149)
Days before Election	0.006***	0.006***	0.006***	0.007***
	(0.001)	(0.001)	(0.001)	(0.001)
2016 election (vs 2012)		2.480***	2.471***	2.449***
		(0.162)	(0.161)	(0.180)
Log sample size			−0.659***	−0.645***
			(0.173)	(0.173)
Internet (vs Phone)				0.175
				(0.212)
Mixed/Other (vs Phone)				−0.419
				(0.272)
Num.Obs.	2436	2436	2436	2436
R2	0.016	0.103	0.108	0.109
R2 Adj.	0.016	0.102	0.107	0.108

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

statistically significant ($p < 0.001$) for models including the predictor. Specifically, the coefficient of 2.48 suggests that polls from the 2016 election had approximately 2.48 percentage points greater absolute error than polls from the 2012 election, holding timing constant. The estimated coefficient changes throughout the models, but remains statistically significant even after accounting for sample size and polling model in Model 4. These findings suggest that difference between election years is able to explain more variation in polling error than timing alone.

Models 3 and 4 extend the model by adding sample size and polling mode respectively. The estimated coefficient for log sample size is -0.659, meaning that doubling the number of respondents is associated with approximately 0.457 percentage points lower absolute error when holding all other predictors constant. On the other hand, polling mode offers minimal additional explanatory value. Relative to polls conducted via phone (reference category), internet polls tend to have approximately 0.175 percentage points greater error, while Mixed/Other polls have 0.419 percentage points less error. However, neither coefficients appear to be statistically significant at the 5% level, meaning there is not enough evidence of difference in error between the different polling modes and Phone polls after adjusting for other predictors.

Overall, polling timing, election year, and sample size indicated statistically significant associations with absolute polling error, while polling model did not. While there was improvements from the initial model (Model 1), the final model (Model 4) only explained 10.936% of the variation in the data, suggesting that majority of the variation remains unexplained by the 4 predictors. Regression diagnostic and assumption checks can be found in Section 4.

4 Statistical appendix

This appendix documents the analysis so the results above are fully reproducible. The annotated code reproduces the model and its assumption checks; the complete project (data, scripts, and commit history) is in the repository linked at the top.

4.1 Data preparation

```
library(readr)
library(tidyverse)
library(dplyr)

# Read cleaned datasets from EDA report
elections <- read_csv("data/1976-2024-president-clean.csv")
polls_2012 <- read_csv("data/state_polls_2012_clean.csv")
polls_2016 <- read_csv("data/state_polls_2016_clean.csv")
```

```r
# Prepare 2012 polling data
# Convert observations into one row per poll
polls_2012_wide <- polls_2012 |>
  filter(candidate %in% c("Obama", "Romney")) |>
  pivot_wider(names_from = candidate, values_from = support) |>
  # New column with election year
  mutate(year = 2012) |>
  # New column with poll margin (Dem - Rep)
  mutate(margin_poll = Obama - Romney)

# Prepare 2016 polling data
# Convert observations into one row per poll
polls_2016_wide <- polls_2016 |>
  filter(candidate %in% c("Clinton", "Trump")) |>
  pivot_wider(names_from = candidate, values_from = support) |>
  # New column with election year
  mutate(year = 2016) |>
  # New column with poll margin (Dem - Rep)
  mutate(margin_poll = Clinton - Trump)

# Combine both election year polling data
polls_2012_2016 <- bind_rows(polls_2012_wide, polls_2016_wide) |>
  # Keep columns necessary for modelling
  select(year, state, poll_id, pollster, end_date, start_date, mode, sample_size, margin_poll

# Check number of polls for each year after combining datasets
polls_2012_2016 |>
  count(year)

# Prepare election results
# Keep election results for only 2012 and 2016, Dem and Rep parties
margin_elections <- elections |>
  filter(year %in% c(2012, 2016)) |>
  # Sum votes across all Rep candidate records and then across all Dem candidate records
  group_by(year, state) |>
  summarise(
    votes_rep = sum(candidatevotes[party_simplified == "REPUBLICAN"],
      na.rm = TRUE
    ),
    votes_dem = sum(candidatevotes[party_simplified == "DEMOCRAT"],
      na.rm = TRUE
```

```
    ),
    # Count total votes in each state for each election year
    votes_total = first(totalvotes),
    .groups = "drop"
  ) |>
  # Calculate election Dem-Rep vote margin as percentage points
  mutate(margin_actual = 100 * (votes_dem - votes_rep) / votes_total)

# Create one dataset containing poll and election results
# Join polling votes with election outcomes
polls_and_elections <- polls_2012_2016 |>
  left_join(margin_elections |> select(year, state, margin_actual),
    by = c("year", "state")
  )
# Check if any polls had no matching election results
sum(is.na(polls_and_elections$margin_actual))

# Prepare regression variables for models
model_data <- polls_and_elections |>
  # Convert polling start and end dates into Date format
  mutate(
    start_date = as.Date(start_date),
    end_date = as.Date(end_date)
  ) |>
  # Add column for election date for the corresponding year
  mutate(election_date = as.Date(if_else(year == 2016, "2016-11-08", "2012-11-06"))) |>
  # Create column with number of days between poll and election
  mutate(days_prior = as.numeric(as.numeric(election_date - end_date))) |>
  # Create column with absolute polling error
  mutate(abs_error = abs(margin_poll - margin_actual)) |>
  # Create column to calculate logarithm of sample size
  mutate(sample_size_log = if_else(sample_size > 0 & !is.na(sample_size),
    log(sample_size),
    NA_real_
  )) |>
  # Election year to be treated as a cateogorical variable for modelling purposes
  mutate(year = factor(year)) |>
  # Polling mode to be condensed into 3 categories and be treated as categorical variable
  mutate(
    mode = case_when(
      mode %in% c(
        "Automated Phone", "Live Phone",
```

```
        "IVR/Live Phone"
      ) ~ "Phone",
      mode == "Internet" ~ "Internet",
      TRUE ~ "Mixed / Other"
    ),
    mode = factor(mode, levels = c("Phone", "Internet", "Mixed / Other"))
  )

# Check model dataset
# Check: all polls have distinct state-year-poll ID
stopifnot(!anyDuplicated(model_data[c("year", "state", "poll_id")]))
# Check: all polls occur before Election day
stopifnot(all(model_data$days_prior >= 0, na.rm = TRUE))
# Check: missing values in variables
model_data |>
  summarise(
    error_miss = sum(is.na(abs_error)),
    year_miss = sum(is.na(year)),
    mode_miss = sum(is.na(mode)),
    sample_miss = sum(is.na(sample_size_log)),
    days_miss = sum(is.na(days_prior))
  )
# Understand distribution of polling modes
model_data |>
  count(year, mode)

# Construct final dataset for models
# Keep observations with no missing values (remove observations with missing sample size)
final_model_data <- model_data |>
  filter(!is.na(sample_size_log))
# Check final number of polls for each year
final_model_data |>
  count(year)
# Glimpse of final model dataset
glimpse(final_model_data)
# Summary of means for main numerical variables
final_model_data |>
  summarise(
    n = n(),
    err_mean = mean(abs_error),
    err_median = median(abs_error),
    days_mean = mean(days_prior),
```

```
    sample_mean = mean(sample_size)
  )

# Save dataset for regressional analysis in data folder
write_csv(final_model_data, "data/final_model_data.csv")
```

4.2 Model

Four OLS regression models were fitted to explore the association between polling characteristics and absolute polling error. The models progressively included timing, election year, log sample size and polling methodology. The comparison of these estimates can be found in Table 3. Table 4 reports the full fitted Model 4 with 95% confidence intervals.

```
# Report fully adjusted model with 95% CI

modelsummary(
  list("Model 4" = model_4),
  statistic = "conf.int",   # 95% CIs beneath each estimate
  coef_map  = c(
    "(Intercept)" = "Intercept",
    "days_prior" = "Days before Election",
    "year2016" = "2016 election (vs 2012)",
    "sample_size_log" = "Log sample size",
    "modeInternet" = "Internet (vs Phone)",
    "modeMixed / Other" = "Mixed/Other (vs Phone)"
  ),
  gof_map   = c("nobs", "r.squared", "adj.r.squared")
)
```

4.3 Assumption checks

Regression diagnostics were carried out for Model 4 to check the aassumptions of linearity, equal variance and approximate normality of residuals. Figure 2 displays residual against fitted values and the normal Q-Q plot for Model 4. An additional plot inspects any potentially influential observations using a Cook's distance plot.

```
# Display 3 plots vertically
par(mfrow = c(3,1), mar = c(4, 4, 2, 1))
# Residual vs fitted plot: checking linearity and equal variance
plot(model_4, which = 1)
```

Table 4: Detailed OLS estimates for Model 4, exploring absolute polling error in 2012 and 2016 elections. Full regression results with 95% confidence intervals for all coefficients. Reference categories is 2012 (election year) and Phone (polling mode).

	Model 4
Intercept	7.597
	[5.344, 9.849]
Days before Election	0.007
	[0.005, 0.009]
2016 election (vs 2012)	2.449
	[2.097, 2.801]
Log sample size	−0.645
	[−0.985, −0.305]
Internet (vs Phone)	0.175
	[−0.239, 0.590]
Mixed/Other (vs Phone)	−0.419
	[−0.951, 0.114]
Num.Obs.	2436
R2	0.109
R2 Adj.	0.108

```r
# Normal Q-Q plot: check normality of residuals
plot(model_4, which = 2)

# Check potentially influential observations
plot(model_4, which = 4)
```

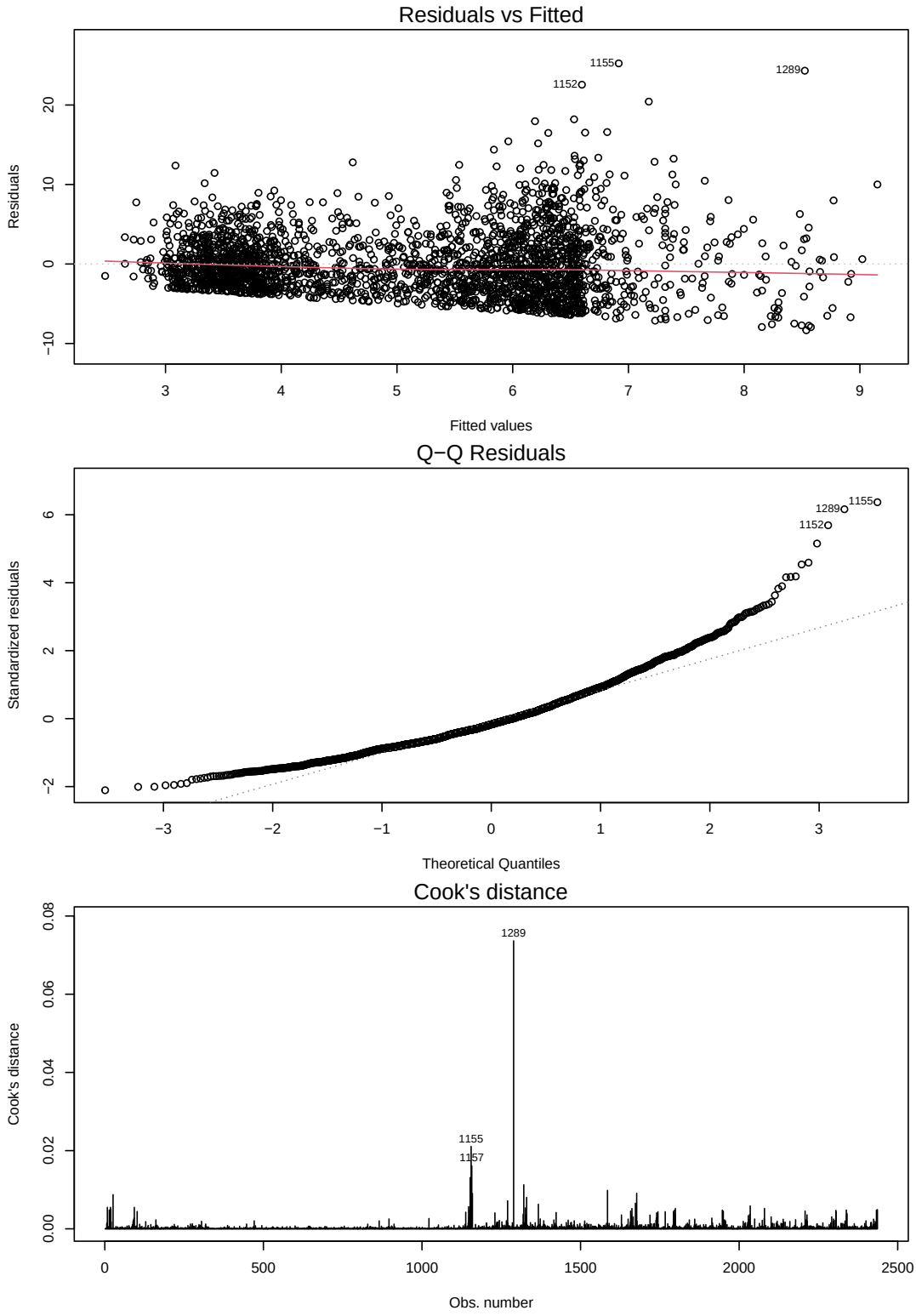


Figure 2: Model 4 diagnostic checks: residual vs fitted values, normal Q-Q plot, and Cook's distance, examining assumptions of the model and any influential observations

The diagnostic plots in Figure 2 showcases some deviations from the assumptions of OLS regression. The residual-vs-fitted plot presents a moderately flat smooth trend, but changes in dispersion may indicate heteroskedasticity. Because of this, standard errors and results may become somewhat unreliable. The normal Q-Q plot also showcases substantial deviation from the reference line, especially in the upper tail, suggesting positively-skewed, non-normal residuals. This is unsurprising as the dependent variable is absolute, which is bounded by 0 but can take large positive values. The Cook's distance plot notes a few potentially influential observations, most evidently observation 1289. These results mean that OLS standard errors and significance tests must be interpreted carefully.

4.4 Computational environment

The R version and packages used for the data preparation, regression models, report creation and quality checks are listed in Table 5 to ensure reproducibility. The final project, consisting of data sets, R scripts and Quarto source code can be found in the GitHub repository linked at the top.

```
pkgs <- c("tidyverse", "modelsummary", "knitr", "lintr")
tibble::tibble(
  Software = c("R", pkgs),
  Version  = c(paste(R.version$major, R.version$minor, sep = "."),
               vapply(pkgs, \(p) as.character(packageVersion(p)), character(1)))
) |>
  kable()
```

Table 5: Core software versions used to produce this report.

Software	Version
R	4.5.1
tidyverse	2.0.0
modelsummary	2.6.0
knitr	1.51
lintr	3.4.0

Kennedy, Courtney, Claire Durand, Kyle McGeeney, Lee Miringoff, Kristen Olson, Douglas Rivers, Lydia Saad, et al. 2018. "An Evaluation of the 2016 Election Polls in the United States." *Public Opinion Quarterly* 82 (1): 1–33. https://doi.org/10.1093/poq/nfx047.

Shirani-Mehr, Houshmand, Sharad Goel, Andrew Gelman, and David Rothschild. 2018. "Disentangling Bias and Variance in Election Polls." *Journal of the American Statistical Association* 113 (522): 607–14.https://doi.org/10.1080/01621459.2018.1448823.